

Introduction to Machine Learning for Text Analysis with Python

Day 4 – Thursday

» Best Practices for Supervised Machine Learning «

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September 17, 2026

Today

Recap

Evaluation Metrics 2

The problem of overfitting

Train/validation/test split

A note on sample size

Exercise: Overfitting

Cross-validation

Finding the optimal (hyper-)parameters

Grid search

Exercise: BERT Cross-validation

More suggestions for improving your models

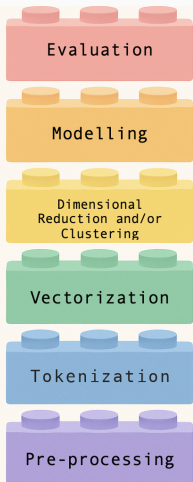
Tuning decision thresholds with ROC curves

Looking forward

Recap

How was yesterday?

Lego pipeline



how to assess the output of your model

how to train a model to describe (UML)
or to predict (SML)

how to group features *or* cases

how to turn text → numbers

how to split text into tokens

how to tidy up text data

Recap: Two big jumps

With respect to the basic Naïve Bayes classifier with a CountVectorizer we make two big jumps this afternoon:

- From BoW to contextual embeddings
- From linear prediction to neural networks

Transfer learning

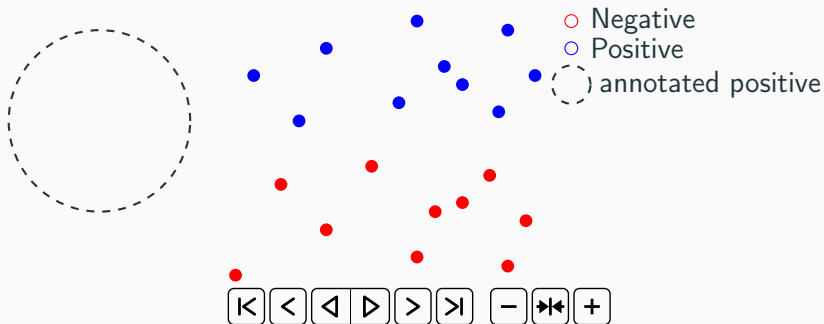
Transfer Learning paradigm

1. **Pre-train** a model on data that is at hand (e.g., Wikipedia, Google News)
2. **Fine-tune** the model on your downstream task (bring in your small-scale annotated dataset)

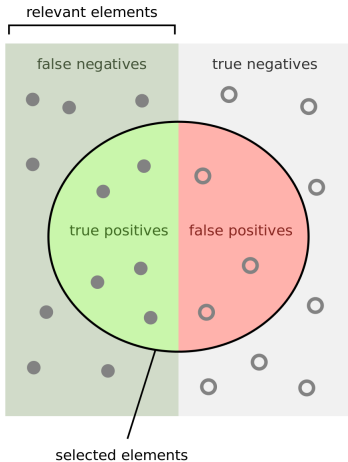
By adding 'task-specific' heads you can produce specific outputs, e.g., classification, text generation, named entity recognition, etc.

Evaluation Metrics 2

Performance: what do we do and how to evaluate it



Points inside the circle are "annotated positive" by the classifier.
Yellow highlight: annotated points (true/false positives/negatives).



Some measures

- Accuracy
- Recall
- Precision
- $F1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$

How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Yesterday's Naïve Bayes baseline model.

	precision	recall	f1-score	support
non-offensive	0.85	0.82	0.83	620
offensive	0.57	0.64	0.60	240
accuracy			0.77	860
macro avg	0.71	0.73	0.72	860
weighted avg	0.78	0.77	0.77	860

Confusion Matrix

	Pred.: non-offensive	Pred.: offensive
non-offensive	506	114
offensive	87	153

Class 1 (Offensive) Metrics:

$$\text{Recall} = \frac{153}{153 + 87} = \frac{153}{240} \approx 0.638$$

$$\text{Precision} = \frac{153}{153 + 114} = \frac{153}{267} \approx 0.573$$

$$F1 = 2 \cdot \frac{0.573 \times 0.638}{0.573 + 0.638} \approx 0.604$$

Stratified Random Baseline

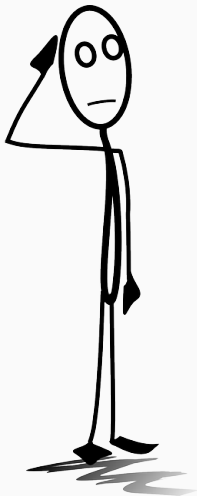
Class Prevalences:

$$P(\text{non-offensive}) = \frac{620}{860} \approx 0.72$$

$$P(\text{offensive}) = \frac{240}{860} \approx 0.28$$

	precision	recall	f1-score	support
non-offensive	0.72	0.72	0.72	620
offensive	0.28	0.28	0.28	240
accuracy			0.60	860
macro avg	0.50	0.50	0.50	860
weighted avg	0.60	0.60	0.60	860

The problem of overfitting



Isn't training multiple models and then selecting the best one a bit like p -hacking?

Yes. We call this problem “overfitting”.

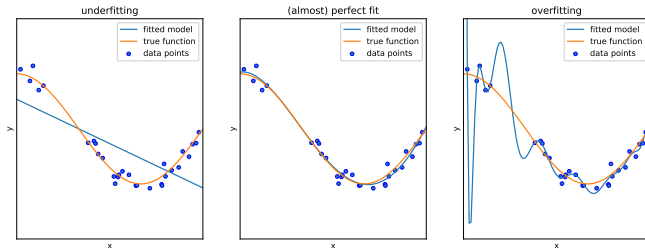


Figure 1: Underfitting and overfitting. Example adapted from https://scikit-learn.org/stable/auto_examples/model_selection/plot_underfitting_overfitting

How to avoid overfitting

1. **Train/test split.** We already do this. It avoids overfitting on the training data.¹
2. **Train/validation/test split.** But maybe we overfit on the test data now? We could set aside *third* dataset.
3. ***k*-fold crossvalidation.** Extending the above such that every case is sometimes $k - 1$ times part of the training data and 1 time part of the validation data.
4. **Regularization.** (in combination with the above) We “penalize” models for being too complex.

¹In classical statistics, the R^2 of an OLS regression is prone to that. Calculating R^2 on a separate test set would be much more conservative.

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Train/validation/test split

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- When you compare a lot of different models (or (hyper-)parameters), you might want to evaluate (compare) them using a third dataset
- e.g., make 80/20 split (train/test); then split first part again 80/20 (train/validation)
- only use the test data *at the very end* to get a final estimate of how good your model is.

In short: Validation data to *select* the best approach; test data to get the accuracy of the approach you chose.

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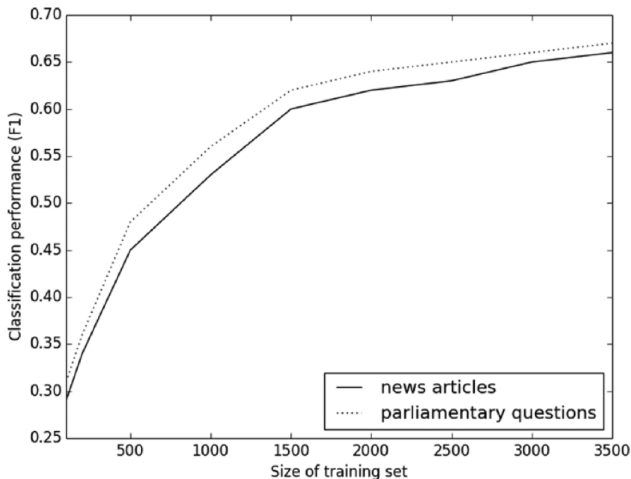
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The problem of overfitting

A note on sample size

Training size vs performance Burscher et al., 2015

FIGURE 1
Learning Curves for the Classification of News Articles and PQs



What does this mean for our research?

It we have 2,000 documents with manually coded frames and topics...

- we can use them to train a SML classifier
- which can code an unlimited number of new documents
- with an acceptable accuracy (at least for some of them)
- Some easier tasks even need only 500 training documents, see Hopkins and King, 2010

→ Which means we can start this class for our Twitter dataset

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Exercise: Overfitting

Cross-validation

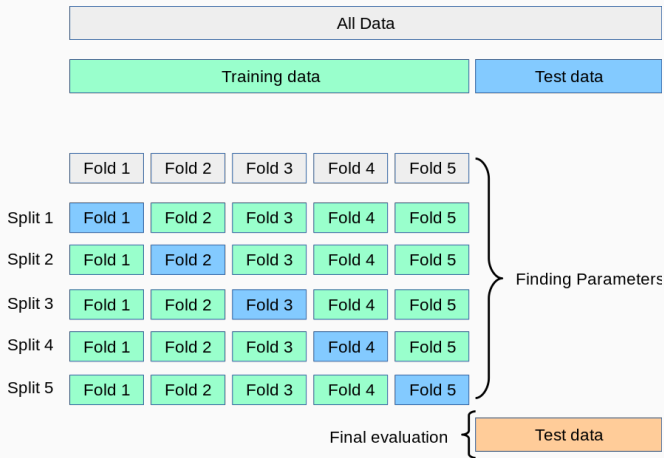


Figure 2: First, we set aside a test dataset for the final evaluation. Then, we split our data into $k = 5$ folds, where each fold is used for validation once (blue) and $k - 1 = 4$ times for training. Source: https://scikit-learn.org/stable/_images/grid_search_cross_validation.png

Cross-validation

```
1 from sklearn.model_selection import cross_val_score
2 from sklearn.naive_bayes import MultinomialNB
3 nb = MultinomialNB() # the classifier we trained this week
4 scores = cross_val_score(nb, train_features, [r[1] for r in reviews], cv
    =10)
5 print(scores)
```

results in:

```
1 [0.858 0.8612 0.8516 0.8528 0.8672 0.8664 0.8576 0.8652 0.8436 0.852 ]
```

We estimate the model 10 times on different training/validation data splits and get 10 different evaluation scores (here: accuracy, but we can use precision, recall, F1, etc. – see examples in the book).

Note that a simple 50:50 train/validation split is identical to setting $k = 2$

Reasons to do Cross-validation

We can get confidence intervals around the scores

- If we have 10 scores instead of one, we can not only get the mean, but also a standard deviation
- If you have two models, both with a mean accuracy (or F1, or whatever) of .85 but one with a large and one with a low standard deviation, you probably prefer the latter – it generalizes better, less likely to suffer from overfitting

Reasons to do Cross-validation

We do not “waste” too much validation data

- If $k = 10$, in each iteration, we use 90% of the data for training.
- We even use *all* data (except test set, of course) for training at least once.
- With train/validation split, we probably need a larger validation set to be sure (e.g., 80/20)
- Especially relevant when annotation is expensive.

Takeaway i

Simple train/test split

- introductory examples/pedagogical reasons
- really small dataset where you cannot afford to set aside validation data
- if – for whatever reason – you do not compare different models and settings.

Takeaway ii

Train/validation/test split

- To compare different model configurations without overfitting on the test data
- Pedagogical reasons, simple start
- Very large datasets or very complex models where setting $k > 2$ would lead to prohibitively expensive computations

Takeaway iii

k -fold crossvalidation

- Best option for comparing multiple models, *especially* when engaging in *hyperparameter tuning* (next section)

In other words: whether you have or create
a train/validation/test or use
cross-validation – just make sure you always
set aside one test set for final evaluation
that you have never used before.

Finding the optimal (hyper-)parameters

hyperparameter a parameter of a model that is not learned through training, but specified in advance

Finding the optimal (hyper-)parameters

Grid search

Hyperparameter optimization with grid search

General idea

Rather than arbitrarily trying some configurations, let's systematically test and compare.

First idea: Use a for-loop! (Example 11.4 in the book)

```
1 configurations =
2 [('NB with Count', CountVectorizer(min_df=5, max_df=.5), MultinomialNB()
3   ),
4   ('NB with TfIdf', TfidfVectorizer(min_df=5, max_df=.5), MultinomialNB())
5   ,
6   ('LogReg with Count', CountVectorizer(min_df=5, max_df=.5),
7     LogisticRegression(solver='liblinear')),
8   ('LogReg with TfIdf', TfidfVectorizer(min_df=5, max_df=.5),
9     LogisticRegression(solver='liblinear'))]
10
11 for description, vectorizer, classifier in configurations:
12     print(description)
13     X_train = vectorizer.fit_transform(reviews_train)
14     X_test = vectorizer.transform(reviews_test)
15     classifier.fit(X_train, y_train)
16     y_pred = classifier.predict(X_test)
17     short_classification_report(y_test, y_pred)
18     print('\n')
```

(where (X_test, y_test) hopefully refers to a validation dataset with another test dataset set aside)

A wishlist

We now tested 2×2 combinations: (NB|LogReg) $\times$ (count|tf·idf).

Wouldn't it be nice if we...

- could simply state this instead of manually creating the list?
- (especially if we had (NB|LogReg) $\times$ (count|tf·idf) $\times$ (min_df=5|min_df=1) $\times$ (max_df=.5|max_df=.8|max_df=.9) = $2 \times 2 \times 2 \times 3 = 24$ options)
- would not have to check manually which one performed best?

That's what scikit-learn's `gridsearch` functionality does! (and the second bullet point is called the *grid*)

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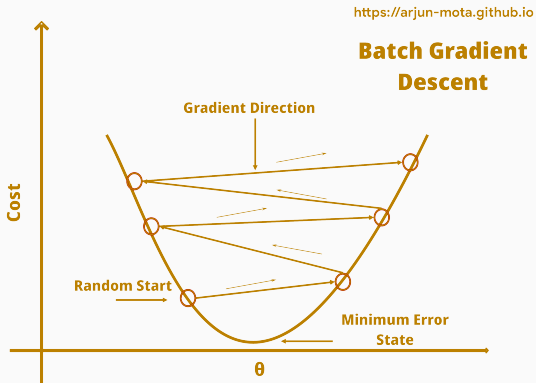
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Gridsearch is especially useful for **hyperparameter optimization**, such as trying out different values for `min_df` and `max_df` on the previous slide.

You could also use a combined approach where you first use a handwritten loop to narrow down the number of candidate models and then tune the model(s) you settled on with a grid search.

BERT: Learning rate



Grid-search takeaway

- When you want to systematically test what happens when you vary a hyperparameter, use grid-search to automatically do so and select the best value
- sometimes already implemented (e.g., `LogisticRegressionCV` as direct replacement for `LogisticRegression`)
- or we write one by hand, such as in today's exercise on cross-validation

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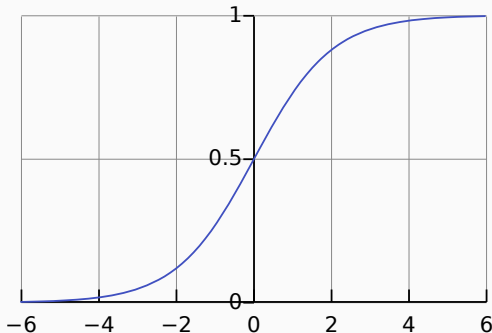
Exercise: BERT Cross-validation

More suggestions

More suggestions

Tuning decision thresholds with ROC curves

From estimate to label



In logistic regression, we use the *sigmoid function* to transform the estimates into probabilities.

To transform the probabilities into binary labels, we use a cutoff (default: 0.5).

Why use 0.5 as cutoff?

- It makes most sense (intuitively, mathematically)
- But remember our precision/recall tradeoff: maybe we want to be 'stricter' or 'less strict'
- Maybe it is important to us that our classifier is balanced and equally good in predicting both classes, even if overall accuracy suffers (slightly)

Let's see what happens if we plot False Positives against True Positives (ROC-curve)

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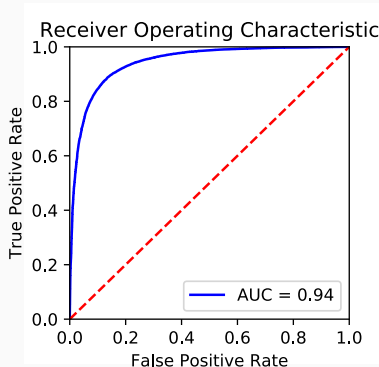
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ROC Curve



- If we choose a threshold such that we get very little false positives, we also get too little true positives.
- Optimum in the upper left corner

Looking forward

Looking forward

And now you...

Project day 1

Try to fill in the blanks for your personal CSS project

Optional: A typical case for
gridsearch - The regularization
parameter C

The regularization parameter λ (or $C = \frac{n}{\lambda}$)

$$\arg \min_{\beta} \left[\sum_{i=1}^n \left(Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|^q \right]$$

We estimate the β -coefficients of a model by optimizing a so-called loss function, i.e minimizing the “cost” that occurs when the prediction is wrong.

If we simply add the sum of the β coefficients to the model, we “punish” large coefficients.

If λ is larger, we punish more.

If $q = 1$, we call it L1 regularization. If we add the squared coefficients ($q = 2$), we call it L2 regularization.

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- scikit-learn uses regularization by default
- (that's why the coefficients of a logistic regression classifier will differ from a "traditional" logistic regression with the statsmodels module)
- Remember that we are *not* interested in a substantive interpretation of the β -coefficients but in a good prediction (which is why "punishing" large coefficients and hence biasing the coefficients is not a problem at all!)
- The penalty will shrink the coefficients towards zero, *which is very useful to avoid overfitting!*
- As always, there is a lot of extra info on the scikit-learn website

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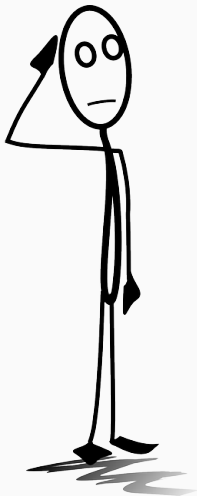
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How harsh should the penalty be?

Hyperparameter optimization with grid search

General idea

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Hyperparameter optimization with grid search

Finding C in a logistic regression using 5-fold cross-validation

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2 logregCV = LogisticRegressionCV(cv=5).fit(train_features, [r[1] for r in
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Hyperparameter optimization with grid search

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2 logregCV = LogisticRegressionCV(cv=5).fit(train_features, [r[1] for r in
    reviews])
```

- Here, we just need to use `LogisticRegressionCV` instead of `LogisticRegression`.
- But we can use it to test any combination of choices

Hyperparameter optimization with grid search

Finding C in a logistic regression using 5-fold cross-validation

```
1 from sklearn.linear_model import LogisticRegressionCV
2 logregCV = LogisticRegressionCV(cv=5).fit(train_features, [r[1] for r in
    reviews])
```

- Here, we just need to use `LogisticRegressionCV` instead of `LogisticRegression`.
- But we can use it to test any combination of choices

```

1 pipeline = Pipeline(steps = [('vectorizer', TfidfVectorizer()), ('
    classifier', LogisticRegression(solver='liblinear'))])
2 grid = {
3     'vectorizer__ngram_range' : [(1,1), (1,2)],
4     'vectorizer__max_df': [0.5, 1.0],
5     'vectorizer__min_df': [0, 5],
6     'classifier__C': [0.01, 1, 100]
7 }
8
9 search = GridSearchCV(estimator=pipeline,
10                       param_grid=grid,
11                       scoring='accuracy',
12                       cv=5,
13                       n_jobs=-1, # use all cpu's
14                       verbose=10)
15 search.fit(reviews_train, y_train)
16 print(f'Using these hyperparameters {search.best_params_}, we get the
    best performance:')
17 print(short_classification_report(y_test, search.predict(reviews_test)))

```

Example 11.6 (book)

Note that the grid is specified as a dictionary where the keys are strings with the name of the step in the pipeline, followed by two underscores, followed by the parameter to tune, and the values are lists of values.

This means that *any* parameter that the classifier or vectorizer takes can be tuned!

The pipeline notation in scikit learn

You might have noticed the Pipeline construct in the last example.

- Machine learning involves multiple steps (e.g., preprocessing → vectorizer → classification)
- We did all of them separately before
- Nothing wrong with that, but to ease use and evaluation of *the whole process* (as needed for the gridsearch), we can define a pipeline.

Example with a pipeline (and add cross-validation)

```
1 from sklearn.feature_extraction.text import TfidfVectorizer
2 from sklearn.linear_model import LogisticRegressionCV
3 from sklearn.pipeline import make_pipeline
4
5 vec = TfidfVectorizer()
6 clf = LogisticRegressionCV()
7 pipe = make_pipeline(vec, clf)
8
9 pipe.fit([r[0] for r in reviews], [r[1] for r in reviews])
10 predictions = pipe.predict([r[0] for r in test])
```

Pipeline takeaway

- In principle, just a different way to write what we already did
- The more steps, the more relevant (e.g., preprocessing → vectorizer → dimensionality-reduction → classification)
- The more you rely on automated evaluation (e.g., grid search) of *multiple* steps in the pipeline, the more useful it is

Some further ideas to look into

Balancing classes

Your classifier probably works better if you have approximately the same amount of annotated training data for both classes (e.g., pos/neg). If getting such data is not an option, you may consider weighing accordingly, e.g. using

```
LogisticRegression(class_weight='balanced')
```

Some further ideas to look into

More advanced pipelines

Consider constructing advanced pipelines, including a dimension reduction step:

https://scikit-learn.org/stable/auto_examples/compose/plot_digits_pipe.html

Some further ideas to look into

Combine different feature sets

E.g, use BOW-features as well as features such as sentence length, number of sentences (or whatever)

<https://scikit-learn.org/stable/modules/generated/sklearn.compose.ColumnTransformer.html>